

Coding & Solution

Date	29 June 2026
Team ID	-
Project Name	Smart lender
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented Smart Lender solution. The project demonstrates a complete machine learning workflow, from data collection and preprocessing to model training, evaluation, and deployment using a Flask web application. The code follows a modular structure with reusable components and proper documentation to ensure maintainability and scalability.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/nandini04112005/Smart_Lender
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Key Features Implemented	Dataset Loading, Data Visualization, Data Preprocessing, Feature Encoding, SMOTE Balancing, Feature Scaling, Train-Test Split, Decision Tree Model, Random Forest Model, KNN Model, Gradient Boosting Model, Cross Validation, Model Saving (.pkl), Flask Web Application, Loan Prediction Interface
Pending / Incomplete Features	None
Setup / Run Instructions	1. Install Vscode. 2. Create a Python environment and install the required libraries (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Flask, Imbalanced-learn, Pickle). 3. Download the loan prediction dataset and place it in the project folder. 4. Run the preprocessing and model training script to generate the trained model (.pkl). 5. Place the saved model file in the project directory. 6. Open the project in Vscode 7. Execute <code>python app.py</code>. 8. Open the localhost URL displayed in the terminal. 9. Enter applicant details on the prediction page and click Submit to view the loan approval prediction

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (if uploaded to GitHub; otherwise change to No)

Additional Notes / Comments:

- The project follows a modular machine learning architecture with separate components for data preprocessing, model training, evaluation, and deployment.
- Four machine learning algorithms (Decision Tree, Random Forest, KNN, and Gradient Boosting) are implemented and compared to select the best-performing model.
- The trained model is serialized using a `.pkl` file and integrated with the Flask web application.
- The web application provides a simple and user-friendly interface for predicting loan approval based on applicant information.
- The project uses clean coding practices, meaningful naming conventions, reusable functions, and structured folders, making it easy to maintain and extend in future versions.